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Spatio-Temporal Graph Neural Networks for Multi-Source Disaggregation and Environmental Impact Assessment of Artificial Nighttime Light Remote Sensing

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ABSTRACT (307 words)

Artificial Nighttime Light (NTL) remote sensing provides indispensable observational data for monitoring human settlement dynamics, socio-economic activity, critical infrastructure resilience, and the adverse ecological impacts of light pollution. However, existing satellite observing systems face significant operational trade-offs between temporal frequency and spatial resolution. While sensors such as the VIIRS Day/Night Band (DNB) deliver daily global observations, their moderate spatial resolution (~500 m) impedes fine-scale characterization of urban lighting morphology and localized ecological disruption. Conversely, sporadic high-resolution nighttime acquisitions (e.g., ISS astronaut photography and commercial cubeSats) lack the continuous radiometric calibration and global revisit cycles necessary for systematic environmental surveillance.

To bridge this critical observational gap, this study introduces an innovative Spatio-Temporal Graph Neural Network (ST-GNN) framework designed for multi-source nighttime light disaggregation and radiometric super-resolution. The proposed architecture models the spatial topology of human settlements and transportation corridors as an interconnected graph, integrating coarse nocturnal radiance from VIIRS-DNB with multi-spectral daytime surface properties acquired by Sentinel-2 (specifically Normalized Difference Built-up Index, Normalized Difference Vegetation Index, and road network geometries). By leveraging graph attention mechanisms and recurrent temporal cells, the model captures non-linear spatial dependencies and seasonal lighting shifts, effectively downscaling NTL radiance to a 10-meter spatial grid with robust cross-sensor radiometric consistency.

The methodology was evaluated across diverse Mediterranean and North African urban-rural gradients, demonstrating superior performance in reconstructing fine-scale artificial illumination patterns while preserving total radiometric energy conservation. The disaggregated nocturnal flux enables precise boundary delineation of skyglow encroachment into protected terrestrial ecosystems, automated detection of localized power outages during extreme events, and rigorous quantification of artificial illumination exposure in proximity to human settlements. By fusing physical spatial constraints with deep geometric learning, this research demonstrates a scalable pathway to enhance the utility of current spaceborne NTL datasets, while offering concrete methodological benchmarks for sensor specifications and mission requirements in prospective ESA nighttime optical observation initiatives.

Keywords: *Artificial Nighttime Light (NTL), Spatio-Temporal Graph Neural Networks, Sentinel-2, Multi-Source Data Fusion, Spatial Disaggregation, Light Pollution & Ecological Impact, Critical Infrastructure Resilience, ESA-ESRIN.*

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